

Drawing App Project Plan

Project goal

Build a browser-based drawing application that demonstrates strong fundamentals (canvas rendering, event handling, state management), plus a small set of standout features suitable for a final submission.

Legend	Difficulty	Rough time (hands-on dev)
	Easy	0.5-2 hours
	Medium	2-6 hours
	Hard	6-12 hours
	Very hard	12+ hours (or multi-session)

Phase 1 - Core MVP

Status	Feature	What it proves / why it matters	Difficulty	Est. time
INCLUDED	Canvas setup + resize	Correct coordinate mapping, retina scaling, resizing without distortion	Easy	1-2h
INCLUDED	Freehand brush tool	Pointer events to strokes, smoothing basics, performance awareness	Easy	2h
COMPLETE	Thickness control (pencil/lines)	UI state to canvas drawing params; consistent tool behavior	Easy	1h
INCLUDED	Color picker	State management + UI integration	Easy	1h
COMPLETE	Opacity control (spray can)	State reset + user affordance	Medium	3-4h
INCLUDED	Export PNG/JPEG	Canvas serialization + download flow	Easy	1-2h

Phase 2 - Tool Variety (Sketch.io-inspired)

Status	Feature	What it proves / why it matters	Difficulty	Est. time
COMPLETE	Eraser tool	Compositing modes or draw-with-background technique	Easy	1h
COMPLETE	Stamps/Shape tools (line/rect/circle)	Preview overlays, drag interactions, geometry	Medium	3-5h
COMPLETE	Color Palette	Custom HSV colour palette with synced native spectrum picker	Medium	2-3h
COMPLETE	Undo / redo	Stacks + command pattern thinking; key for usability	Medium	3-4h
INCLUDED	Mirror drawing	Coordinate transforms and symmetry logic	Medium	2-4h

PLANNED	Fill tool (bucket)	Flood-fill algorithm + performance constraints	Hard	5-7h
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Phase 3 - Design/US and Polish for Final

Status	Feature	What it proves / why it matters	Difficulty	Est. time
COMPLETED	Toolbar with icons	Information architecture + clean UX	Easy	2h
COMPLETED	Keyboard shortcuts	Event handling + accessibility	Medium	2h
COMPLETED	Cursor preview (brush size)	Real-time feedback loop; polish	Medium	2-3h
COMPLETED	Responsive layout	App layout that works on laptop/tablet	Medium	3h

Phase 4 - 1-2 Impressive Extensions to Choose From

Status	Feature	What it proves / why it matters	Difficulty	Est. time
COMPLETE	Image import (draw over photo)	File APIs + image scaling + non-destructive workflow	Medium	2-3h
PLANNED	Zoom + pan	Canvas transforms + hit testing correctness	Hard	6-8h
PLANNED	Layers (basic)	Compositing strategy; per-layer bitmap + ordering	Hard	8-12h
PLANNED	Gradient & Pattern Fills	color blending, coordinate-based rendering, and reusable drawing abstractions	Hard	3-4h
COMPLETE	Project save/load	Serialize state (layers/tools) to JSON; localStorage/files	Medium	3-5h

Phase 5 - One 'Wow' Feature for the Final

Status	Feature	What it proves / why it matters	Difficulty	Est. time
PLANNED	Time-lapse replay	Store stroke events + playback; great demo feature	Hard	6-10h
PLANNED	Symmetry modes (multi-axis)	Generalize mirror logic (radial, vertical + horizontal)	Hard	6-10h
COMPLETE	Custom brushes (patterns/textures)	Stamping + performance; optional WebGL later	Hard	6-8h

Scope for Drawing App Final

Undo/Redo + Shapes + Export + Interactive Toolbar (cursor preview) + one standout (either Fill tool).

Notes (inspired by sketch.io without copying)

Sketch.io feels powerful because it combines tool variety, layers/organization, and easy exporting. Your best ROI is: (1) fast tool switching, (2) undo/redo reliability, and (3) one memorable advanced feature.

Brain Storming

1. Hover over the color palette icon for 0.5 seconds → shows a small preview of current color
2. Click on the preview → expands to show a larger preview swatch and a spectrum color picker
3. Use the spectrum picker to select any color
4. The preview swatch updates to show the selected color
5. Mouse away hides the palette after 1 second delay
6. Append the thickness and opacity slider to the color palette
- 7.

Brain Storming

Tools

Line Tool

- Add opacity slider alongside the existing thickness slider
- Apply opacity to the stroke color using the color function with main channel

Spray Can Tool

- Add opacity slider alongside the existing thickness slider
- Apply opacity to the stroke color using the color function with main channel

Drawing Tool

- Add opacity slider alongside the existing thickness slider
- Apply opacity to the stroke color using the color function with main channel

Mirror Tool

- Add opacity slider alongside the existing thickness slider
- Apply opacity to the stroke color using the color function with main channel

Stamp Tool

- Add opacity slider alongside the existing thickness slider
- Apply opacity to the stroke color using the color function with main channel
- Add floating panel when user hovers over the icon to preview all available stamps
- Preview of floating stamps also has scrolling capabilities (when there are a large amount of stickers)
- Add search engine for stickers based on category
- Cursor image reflects Stamp selected and current size of the stamp

Color Palette Tool

- Add floating panel when user hovers over the icon to preview current color selected
- Add HSV palette with ability to utilize cursor to adjust hex RGB value
- Append Opacity and Thickness slider to the floating color palette tool
- Set/store opacity and thickness slider value globally (ei. to prevent users from having to continuously adjust settings)
- Add general color palette blocks with an array of custom selected colors (around 30-45 colors)

Abstract Paint Brushes Tool

- Add opacity slider alongside the existing thickness slider
- Apply opacity to the stroke color using the color function with main channel
- Add/Design different patterns of lines/shapes to create the illusion of different brush strokes
- Add floating panel when user hovers over the icon to preview all available brush strokes

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